

SATQUERY-AI MISSION REPORT

GEOSPATIAL BI-TEMPORAL CHANGE DETECTION & EVIDENCE REPORT • SMART INDIA HACKATHON

Mission ID:	4036c430-e3d3-41a5-9ef4-3b25b40722fb	Generated:	2026-09-08T19:52:42
Mode:	bi_temporal	Detector Threshold:	0.70 (LEVIR-CD Calibrated)
Checkpoint:	best_model.pt (SiameseUNet)	Confidence:	N/A — Uncalibrated

1. EXECUTIVE SUMMARY

SatQuery-AI completed bi-temporal change analysis for mission ID 4036c430-e3d3-41a5-9ef4-3b25b40722fb. The Siamese U-Net detector identified 9.09% changed pixels (71,495 of 786,432 total pixels) between the supplied acquisitions.

2. VISUAL EVIDENCE & SENSOR APERTURE

T1: Before Acquisition



T2: After Acquisition



Siamese U-Net Change Heatmap



3. QUANTITATIVE CHANGE DETECTION & SPATIAL ANALYTICS

Total Scene Pixels	786,432	Changed Pixels	71,495 (9.09%)
Unchanged Pixels	714,937 (90.91%)	Model Threshold	0.70
Coordinate System (CRS)	Not available	Spatial Resolution (GSD)	Not available
Physical Changed Area	Unavailable (No reliable GSD)	Total Physical Scene	Unavailable (No reliable GSD)

4. CONNECTED COMPONENT HOTSPOTS (TOP LOCALIZED CLUSTERS)

Cluster #	Pixel Area	Share of Change	Centroid (px: X, Y)	Bounding Box (px: x1, y1, x2, y2)
Cluster 6	38,133 px	53.34%	(727.0, 314.2)	[0, 0, 0, 0]
Cluster 2	12,269 px	17.16%	(630.7, 162.2)	[0, 0, 0, 0]
Cluster 51	5,826 px	8.15%	(886.5, 459.2)	[0, 0, 0, 0]
Cluster 35	4,040 px	5.65%	(834.0, 391.1)	[0, 0, 0, 0]
Cluster 60	1,721 px	2.41%	(622.0, 496.3)	[0, 0, 0, 0]

5. REGION OF INTEREST (ROI) INVESTIGATION

ROI investigation was not performed.

6. AI VISION-LANGUAGE INTERPRETATION (CHANGE VQA)

Status: Executed via Domain-Adapted VLM	Confidence: N/A — Uncalibrated
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[illegible][illegible]

Step	Title	Status	Execution Detail
step-1	Query Received	DONE	Query accepted: "Bi-temporal satellite change detection assessment" Mode: bi_temporal Images: 2

step-2	Input Validation	DONE	optical_before.jpg: JPEG Modality=rgb_optical (75%) file_exists=✓ (123512 B) 1024×768px bands=3 ; optical_after.jpg: JPEG Modality=rgb_optical (75%) file_exists=✓ (123605 B) 1024×768px bands=3 ; Pair Compatibility: COMPATIBLE ; Temporal: unknown
step-3	Task Classification	DONE	Detected intents: [grounding: 1.00] [change_detection: 2.00] [change_description: 1.40] Tasks: grounding, change_detection, change_description
step-4	Tool Selection	DONE	change_detector[REAL] → Bi-temporal Change Detector 0.3.0-p0 change_vqa[REAL] → Change-VQA Language Model 0.3.0-p0 spatial_analyzer[MOCK] → Spatial Analyzer 0.1.0-mock
step-5	Parameters	DONE	change_detector: algorithm=siamese-unet-model, tile_size=256, tile_overlap=32, threshold=0.7 change_vqa: temperature=0.2, max_tokens=768, provider=local spatial_analyzer: compute_areas=True, proximity_analysis=True
step-6	Processing	DONE	Bi-temporal Change Detector[REAL]: 730ms Change-VQA Language Model[REAL]: 28325ms Spatial Analyzer[MOCK]: 120ms real SiameseUNet model change map generated (LEVIR-CD trained checkpoint)
step-7	Aggregation	DONE	Merging tool outputs Concatenating answer sections Confidence: null (no calibrated model-reported score)
step-8	Completion	DONE	Analysis complete Confidence: null Total elapsed: 29.14s tools=[change_detector=real, change_vqa=real, spatial_analyzer=mock]

8. SENSITIVITY, UNCALIBRATED METRICS & LIMITATIONS

- Confidence is not calibrated across detector and vision-language models (strictly reported as N/A — Uncalibrated).
- Change detection operates on pixel-level contrast via Siamese U-Net without assigning semantic land-cover categories.
- Physical area unavailable — reliable spatial resolution (GSD) metadata was not provided in image headers.
- Geospatial coordinates unavailable — analysis spatial coordinates are reported strictly in pixel space.
- ROI investigation was not performed for this report compilation.